The neutral platinum complex *cis*-[Pt(NH_3)$_2$$Cl_4$] consists of a platinum(IV) atom with coordinate covalent bonds to four negatively charged $Cl^–$ ligands and two neutral NH_3 ligands that are positioned on the same side of the Pt atom. The *cis*-[Pt(NH_3)$_2$$Cl_4$] complex acts as a prodrug by converting into the chemotherapy drug cisplatin (*cis*-[Pt(NH_3)$_2$$Cl_2$]) through an oxidation-reduction reaction with thiols present in the cytosol of a cell. A scientist studying the reaction between *cis*-[Pt(NH_3)$_2$$Cl_4$] and the thioglycolate anion (an anionic thiol) proposes that the reaction occurs by the two-step mechanism shown in Figure 1.

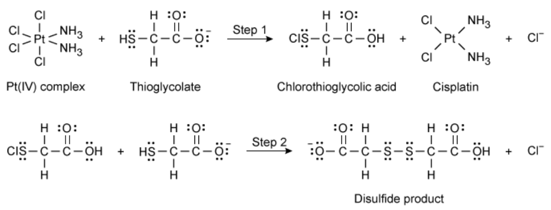


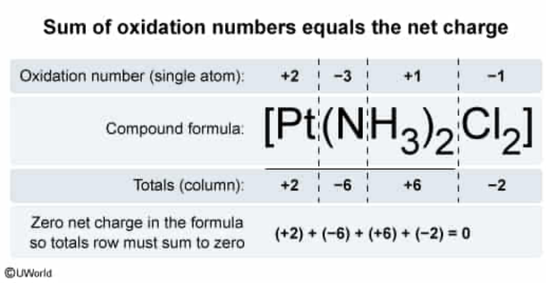
Figure 1 Proposed mechanism for the reaction of *cis*-[Pt(NH_3)$_2$$Cl_4$] with thioglycolate

In cisplatin, the oxidation number of platinum is:

- A. 0 [16%]
- ✔ B. +2 [53%]
- C. +4 [25%]
- D. +6 [5%]

Correct
53% answered correctly

Explanation:



Oxidation numbers account for the gain or loss of electrons by an atom relative to the number of electrons present in its elemental state. Oxidation numbers may be assessed using Lewis structures. However, the values may be alternatively assessed from the compound formula using oxidation state rules. By this second method, atoms with consistent oxidation states (ie, those that rarely vary) are used to determine oxidation states for atoms that have variable states (eg, transition metals).

Oxidation numbers differ from the formal charge in that oxidation numbers account for electron changes due to oxidation and reduction, whereas formal charge accounts for distribution of charge among atoms. In any molecule, however, the **sum of the oxidation states** of every atom in the structure must equal the **net charge** of the compound formula.

Using the **oxidation state rules** for the formula *cis*-$[Pt(NH_3)_2Cl_2]$ (ie, cisplatin) shows that platinum has an oxidation state of +2. Although platinum is a transition metal that has a variable oxidation state (depending on the compound formed), the oxidation states of N, H, and Cl are more consistent and can be used to find the value for platinum. Because *cis*-$[Pt(NH_3)_2Cl_2]$ has no net charge, the oxidations states of all atoms — the Pt atom, two nitrogen atoms (each with a state of −3), six hydrogen atoms (each with a state of +1), and two chlorine atoms (each with a state of −1) — must sum to 0.

$$[Pt(NH_3)_2Cl_2] = 0 \text{ net charge}$$

$$Pt + 2(-3) + 6(+1) + 2(-1) = 0$$

Solving for Pt gives its oxidation number in the compound:

$$Pt - 6 + 6 - 2 = 0$$

$$Pt = +2$$

(Choice A) A value of 0 is the net charge of the compound formula.

(Choice C) A value of +4 is the oxidation number of the Pt atom in *cis*-$[Pt(NH_3)_2Cl_4]$ (ie, the Pt(IV) complex), not the oxidation number of the Pt atom in *cis*-$[Pt(NH_3)_2Cl_2]$ (ie, cisplatin).

(Choice D) A value of 6 is the coordination number of the Pt atom in *cis*-$[Pt(NH_3)_2Cl_4]$, not the oxidation number of the Pt atom.

Educational objective:
The oxidation number of each atom in a compound formula may be determined using oxidation state rules and recognizing that the sum of the oxidation states of each atom must equal the net charge of the compound or ion.

Time spent:0 sec
Subject:
General Chemistry

QID:403578
System:
Interactions of Chemical Substances

The neutral platinum complex *cis*-[Pt(NH_3)$_2$$Cl_4$] consists of a platinum(IV) atom with coordinate covalent bonds to four negatively charged Cl^- ligands and two neutral NH_3 ligands that are positioned on the same side of the Pt atom. The *cis*-[Pt(NH_3)$_2$$Cl_4$] complex acts as a prodrug by converting into the chemotherapy drug cisplatin (*cis*-[Pt(NH_3)$_2$$Cl_2$]) through an oxidation-reduction reaction with thiols present in the cytosol of a cell. A scientist studying the reaction between *cis*-[Pt(NH_3)$_2$$Cl_4$] and the thioglycolate anion (an anionic thiol) proposes that the reaction occurs by the two-step mechanism shown in Figure 1.

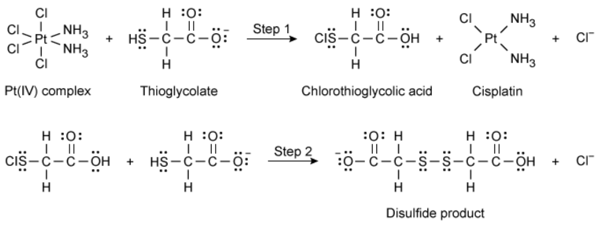


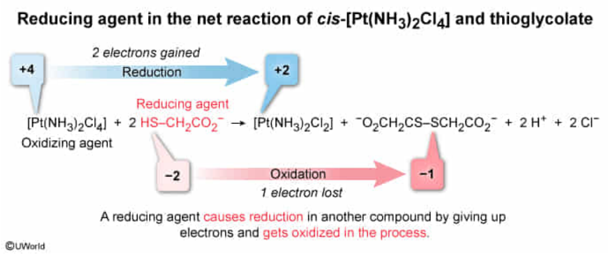
Figure 1 Proposed mechanism for the reaction of *cis*-[Pt(NH_3)$_2$$Cl_4$] with thioglycolate

Which chemical species acts as the reducing agent in the net reaction from the two-step mechanism in Figure 1?

- A. *cis*-[Pt(NH_3)$_2$$Cl_4$] [26%]
- ✔ B. Thioglycolate [59%]
- C. Chlorothioglycolic acid [10%]
- D. Cisplatin [3%]

Correct
59% answered correctly

Explanation:



In a redox reaction, oxidation and reduction always occur simultaneously. An atom is reduced by gaining one or more electrons from another atom that loses those electrons (ie, gets oxidized). To **track electron transfers** between atoms during reactions, an **oxidation number** is assessed for each atom in a compound relative to the oxidation state of a neutral atom of that element.

Comparing the oxidation number of an element in the reactants with the oxidation number of that same element in the products indicates whether the element has gained or lost electrons. An *increased* oxidation number indicates a *loss* of electrons (oxidation), whereas a *decreased* oxidation number signals a *gain* of electrons (reduction). A change of **one unit** occurs **for each electron** transferred.

The reducing agent in a reaction is the chemical species that *causes reduction* in another atom by giving up electrons (ie, the reducing agent itself gets oxidized in the process). Conversely, the **oxidizing agent** is the chemical species that *causes oxidation* by taking electrons from (ie, oxidizing) another atom and getting reduced in the process.

The balanced net reaction for the mechanism in Figure 1 is:

$$[Pt(NH_3)_2Cl_4] + 2\ HS\text{–}CH_2CO_2^- \rightarrow [Pt(NH_3)_2Cl_2] + {}^-O_2CH_2CS\text{–}SCH_2CO_2^- + 2\ H^+ + 2\ Cl^-$$

Assessing and comparing oxidation states for the atoms in the reaction shows that the S atom in thioglycolate ($HS\text{–}CH_2CO_2^-$) is oxidized from −2 to −1. The electrons lost by thioglycolate causes the Pt atom to be reduced from +4 to +2. Therefore, thioglycolate *causes* reduction and is the reducing agent in the net reaction.

(Choice A) *Cis*-$[Pt(NH_3)_2Cl_4]$ is the *oxidizing agent* in the net reaction.

(Choice C) Chlorothioglycolic acid is an intermediate and is not part of the net reaction. In addition, the S atom in the Cl–S bond gets reduced in Step 2 of the mechanism. A reducing agent itself must get *oxidized* due to its action.

(Choice D) Cisplatin is a *product* of the net reaction, so it cannot be the reducing agent.

Educational objective:
A reducing agent causes a reduction in another compound by giving electrons to the other compound. Therefore, a reducing agent itself undergoes oxidation due to the loss of electrons. Oxidation numbers track the gain or loss of electrons by atoms in a reaction.

Time spent: 0 sec
Subject: General Chemistry

QID: 403579
System: Interactions of Chemical Substances

The neutral platinum complex *cis*-[Pt(NH_3)$_2$$Cl_4$] consists of a platinum(IV) atom with coordinate covalent bonds to four negatively charged Cl^- ligands and two neutral NH_3 ligands that are positioned on the same side of the Pt atom. The *cis*-[Pt(NH_3)$_2$$Cl_4$] complex acts as a prodrug by converting into the chemotherapy drug cisplatin (*cis*-[Pt(NH_3)$_2$$Cl_2$]) through an oxidation-reduction reaction with thiols present in the cytosol of a cell. A scientist studying the reaction between *cis*-[Pt(NH_3)$_2$$Cl_4$] and the thioglycolate anion (an anionic thiol) proposes that the reaction occurs by the two-step mechanism shown in Figure 1.

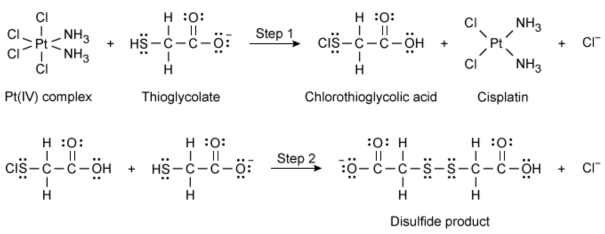


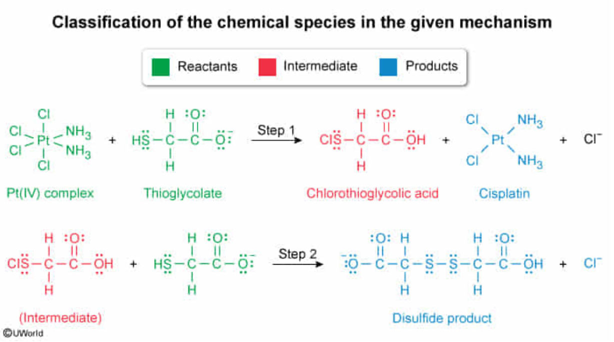
Figure 1 Proposed mechanism for the reaction of *cis*-[Pt(NH_3)$_2$$Cl_4$] with thioglycolate

Which chemical species is an intermediate in the net reaction from the two-step mechanism in Figure 1?

- A. Cisplatin [13%]
- B. *cis*-[Pt(NH_3)$_2$$Cl_4$ [1%]
- C. Thioglycolate [3%]
- D. Chlorothioglycolic acid [81%]

Correct
81% answered correctly

Explanation:



A reaction mechanism is a sequence of elementary reaction steps that yield an overall (net) reaction. Mechanisms include the reactants and any intermediates or catalysts involved in each step, and the mechanism demonstrates how the resulting products of the net reaction are formed. An intermediate is a transient chemical species that is formed in one elementary step and consumed in a subsequent step. As such, intermediates do not appear in the net reaction.

In the mechanism in Figure 1, chlorothioglycolic acid is formed in Step 1 and then subsequently consumed in Step 2. Therefore, chlorothioglycolic acid is an intermediate in the reaction mechanism and does not appear in the net reaction.

(Choice A) Although cisplatin is formed in Step 1, it is not consumed in Step 2. Consequently, cisplatin is a product in the net reaction and is not an intermediate.

(Choice B) *Cis*-$[Pt(NH_3)_2Cl_4]$ is a reactant in both Step 1 and in the net reaction because it is not produced in a prior elementary step.

(Choice C) Although thioglycolate is a reactant in both Steps 1 and 2, thioglycolate is not an intermediate because it is present at the onset of the reaction and is not produced in a prior step.

Educational objective:
An intermediate is a transient chemical species that is formed in one elementary step and consumed in a subsequent step. Intermediates do not appear as reactants or products in the overall (net) reaction.

Time spent:0 sec
Subject:
General Chemistry

QID:403580
System:
Interactions of Chemical Substances

The neutral platinum complex *cis*-[$Pt(NH_3)_2Cl_4$] consists of a platinum(IV) atom with coordinate covalent bonds to four negatively charged Cl^- ligands and two neutral NH_3 ligands that are positioned on the same side of the Pt atom. The *cis*-[$Pt(NH_3)_2Cl_4$] complex acts as a prodrug by converting into the chemotherapy drug cisplatin (*cis*-[$Pt(NH_3)_2Cl_2$]) through an oxidation-reduction reaction with thiols present in the cytosol of a cell. A scientist studying the reaction between *cis*-[$Pt(NH_3)_2Cl_4$] and the thioglycolate anion (an anionic thiol) proposes that the reaction occurs by the two-step mechanism shown in Figure 1.

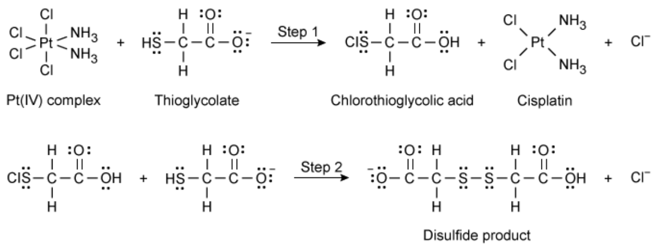


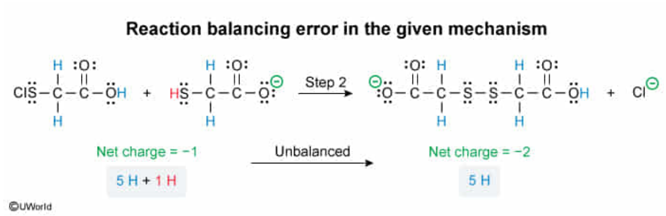
Figure 1 Proposed mechanism for the reaction of *cis*-[$Pt(NH_3)_2Cl_4$] with thioglycolate

After reevaluating the proposed mechanism in Figure 1, the scientist studying the reaction concludes that the mechanism contains an error in the reaction balancing. Is this conclusion correct?

- A. Yes; the net reaction from the mechanism is unbalanced only in terms of the number of atoms. [21%]
- B. Yes; the net reaction from the mechanism is unbalanced only in terms of the net charge. [25%]
- C. Yes; the net reaction from the mechanism is unbalanced in terms of both the number of atoms and the net charge. [41%]
- D. No; the net reaction from the mechanism is balanced, and the scientist is mistaken about the error. [10%]

Correct
42% answered correctly

Explanation:



Due to the **law of conservation of mass**, a **balanced reaction** should have the same number of atoms of each type **on both sides** of the reaction arrow. Likewise, the positively charged protons and negatively charged electrons within the atoms are conserved. Consequently, the same net charge must be present **on both sides** of the equation.

In the given mechanism, Step 1 is balanced in terms of the number of atoms and net charge, but Step 2 is not balanced in either. In Step 2, the H atom from the H–S bond in the thioglycolate reactant is not accounted for in the products. This also produces an imbalance in the net charge, giving the reactants side a net charge of −1 and the products side a net charge of −2. Because Step 2 is not balanced, the net reaction from Steps 1 and 2 will also be unbalanced.

Therefore, the conclusion of the scientist is correct, and the net reaction from the mechanism is unbalanced in terms of both the number of atoms and net charge.

Although not required to answer the question, the given mechanism can be corrected by accounting for the H atoms from the S–H bonds broken in both steps. Consequently, the corrected and balanced mechanism and its net reaction are as follows:

Step 1:	$[Pt(NH_3)_2Cl_4]$ + $\mathbf{HS}\text{–}CH_2CO_2^-$	$\rightarrow [Pt(NH_3)_2Cl_2]$ + $\cancel{ClS\text{–}CH_2CO_2H}$ + Cl^-
Step 2:	$\cancel{ClS\text{–}CH_2CO_2H}$ + $\mathbf{HS}\text{–}CH_2CO_2^-$	$\rightarrow HO_2CH_2CS\text{–}SCH_2CO_2\mathbf{H}$ + Cl^- (*corrected reaction*)
Net:	$[Pt(NH_3)_2Cl_4]$ + 2 $\mathbf{HS}\text{–}CH_2CO_2^-$	$\rightarrow [Pt(NH_3)_2Cl_2]$ + $\mathbf{H}O_2CH_2CS\text{–}SCH_2CO_2\mathbf{H}$ + 2 Cl^-

(Choices A, B, and D) Although Step 1 is fully balanced, Step 2 is unbalanced in both the number of atoms and net charge. This will yield an unbalanced net reaction for the mechanism as given. Accordingly, the scientist is correct in recognizing an error.

Educational objective:

Based on the law of conservation of mass, the atoms and their constituent protons and electrons are conserved in a chemical reaction. As such, a balanced reaction must have the same number of each type of atom and the same total charge on both sides of the reaction arrow.

Time spent: 0 sec	**QID**:403581
Subject:	**System:**
General Chemistry	Interactions of Chemical Substances

The neutral platinum complex *cis*-[$Pt(NH_3)_2Cl_4$] consists of a platinum(IV) atom with coordinate covalent bonds to four negatively charged Cl^- ligands and two neutral NH_3 ligands that are positioned on the same side of the Pt atom. The *cis*-[$Pt(NH_3)_2Cl_4$] complex acts as a prodrug by converting into the chemotherapy drug cisplatin (*cis*-[$Pt(NH_3)_2Cl_2$]) through an oxidation-reduction reaction with thiols present in the cytosol of a cell. A scientist studying the reaction between *cis*-[$Pt(NH_3)_2Cl_4$] and the thioglycolate anion (an anionic thiol) proposes that the reaction occurs by the two-step mechanism shown in Figure 1.

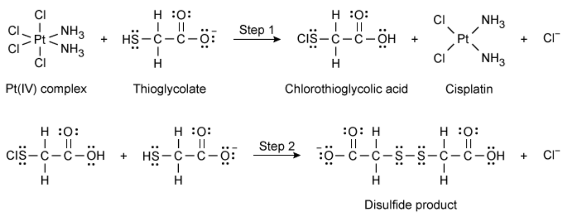


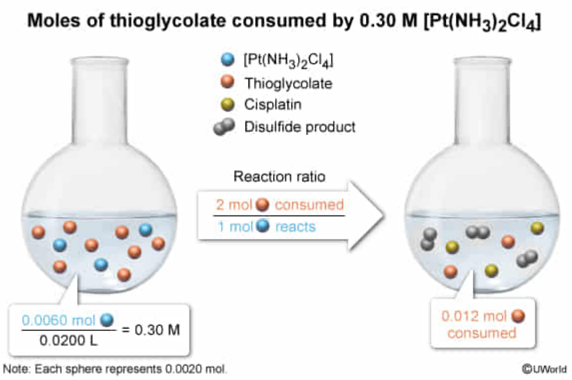
Figure 1 Proposed mechanism for the reaction of *cis*-[$Pt(NH_3)_2Cl_4$] with thioglycolate

If 20.0 mL of a solution containing 0.30 M [$Pt(NH_3)_2Cl_4$] reacts according to the two-step mechanism in Figure 1, what is the total number of moles of thioglycolate will be consumed in the net reaction resulting from both steps? (Note: Assume that the platinum complex is the limiting reactant and that the reaction goes to completion.)

- A. 3.0×10^{-3} mol [6%]
- B. 6.0×10^{-3} mol [38%]
- C. 1.2×10^{-2} mol [49%]
- D. 6.0×10^{-1} mol [6%]

Correct
49% answered correctly

Explanation:



The molar concentration (molarity) of a chemical species is defined as the **number of**

moles of the species **per liter of solution (mol/L)**. As such, if a solution has a concentration of 0.30 M $[Pt(NH_3)_2Cl_4]$, the number of moles of $[Pt(NH_3)_2Cl_4]$ present in 20.0 mL (0.0200 L) is:

$$0.0200\ \cancel{\text{L}} \times \frac{0.30\ \text{mol}\left[\text{Pt}\left(\text{NH}_3\right)_2\text{Cl}_4\right]}{1\ \cancel{\text{L}}} = 0.0060\ \text{mol}\left[\text{Pt}\left(\text{NH}_3\right)_2\text{Cl}_4\right]$$

In any chemical reaction, the moles of one chemical species can be related to the moles of another chemical species by the mole ratios (ie, stoichiometry) from a chemical reaction or compound formula. Figure 1 shows that each time the mechanism is completed, 2 moles of thioglycolate ($HS–CH_2CO_2^-$) are consumed (ie, 1 mol in each step) for every 1 mole of $[Pt(NH_3)_2Cl_4]$ that reacts. Therefore, if $[Pt(NH_3)_2Cl_4]$ is the limiting reactant and the reaction goes to completion, the number of moles of thioglycolate consumed in the reaction is:

$$0.0060\ \cancel{\text{mol}\left[\text{Pt}\left(\text{NH}_3\right)_2\text{Cl}_4\right]} \times \frac{2\ \text{mol HS} — \text{CH}_2\text{CO}_2^-}{1\ \cancel{\text{mol}\left[\text{Pt}\left(\text{NH}_3\right)_2\text{Cl}_4\right]}} = 0.012\ \text{mol HS} — \text{CH}_2\text{CO}_2^- \implies 1.2 \times 10^{-2}\ \text{mol}$$

(Choice A) A quantity of 3.0×10^{-3} mol is calculated by mistakenly inverting the mole ratio (ie, dividing by 2 instead of multiplying by 2) during the second step of the calculation.

(Choice B) A quantity of 6.0×10^{-3} mol is calculated by incorrectly assuming that only 1 mole of $HS–CH_2CO_2^-$ is consumed for each mole of $[Pt(NH_3)_2Cl_4]$ that reacts (ie, using a 1:1 mole ratio during the second step of the calculation).

(Choice D) A quantity of 6.0×10^{-1} mol (with incorrect units) is found by mistaking the molarity (mol/L) for moles and failing to multiply the 0.30 M concentration by the volume of the solution (ie, the first step of the calculation).

Educational objective:
The number of moles present in a solution can be calculated by multiplying the molarity of the solution by its volume. Mole ratios from a reaction mechanism can be used to relate the moles of one chemical species to the moles of another species in a reaction calculation.

Time spent:0 sec
Subject:
General Chemistry

QID:403582
System:
Interactions of Chemical Substances